

From Roadmap to Mastery Graph: An Open-Source Framework for Self-Directed AI Education

This report details the design of a comprehensive, modular, competency-based learning framework for mastering artificial intelligence. It is tailored for a self-directed learner utilizing only freely available online resources, replacing a rigid timeline with a flexible, fluid system. The framework is built upon three core pillars: a modular curriculum structured as a skill graph, robust competency assessment mechanisms, and an automated analytics engine that provides actionable feedback. This design directly addresses the user's goal of creating a full learning system that supports goal planning, progress recording, and curriculum navigation across all essential AI knowledge areas, from foundational mathematics to advanced systems design and ethical considerations.

Architecting the Learning System: From Timeline to Skill Graph

The fundamental shift required by this research goal is the replacement of a fixed, linear timeline with a dynamic, competency-based architecture . This approach moves beyond measuring progress by elapsed time and instead focuses on demonstrable skill acquisition, a principle deeply rooted in educational theory. The proposed framework is not merely a collection of resources but a cohesive system designed around the concept of a "Skill Graph," a model that represents knowledge domains as interconnected nodes with defined dependencies [129132](#). This structure is the cornerstone of enabling true personalization and adaptive learning, allowing the system to guide a learner along a unique path toward mastery rather than forcing them onto a single, predetermined route. The architectural blueprint is therefore conceived as a three-layered system: the Content & Curriculum Layer, the Assessment & Analytics Layer, and the User Interface & Experience Layer. This separation allows for modularity and future extensibility, aligning with the user's explicit request for a modular design .

The theoretical underpinnings for this shift are strong. Mastery learning, an educational strategy where students advance only after demonstrating proficiency in prerequisite material, has been shown to significantly improve student achievement [66](#). A meta-analysis of 108 studies found that mastery learning programs raised final exam scores by approximately half a standard deviation, with lower-aptitude students experiencing even greater gains, suggesting it helps close performance gaps [66](#). A four-year action research study at a Latin American university transformed an introductory computer science course into a modular mastery-based format, resulting in a dramatic increase in the pass rate from 19.3% to 77.1% and a reduction in attrition from 25.0% to 3.8% [67](#). The core philosophy posits that nearly all students can master a skill given sufficient time, appropriate instruction, and support, which stands in contrast to traditional "one size fits all" models that often leave struggling learners behind [34](#) [67](#). However, a critical challenge identified in implementing such systems is procrastination; the same CS1 course study noted that without prescribed deadlines for assessments, many students delayed their work until the end of the semester, hindering consistent progress [66](#) [67](#). Therefore, the framework must incorporate motivational scaffolding, such as setting aspirational milestones or using analytics to flag learners who are falling behind their own potential pace, to mitigate this risk.

The central innovation of this proposed architecture is the adoption of a Skill Graph to model the curriculum. Instead of a simple list of courses or topics, the curriculum is represented as a network where skills are nodes and dependencies are directed edges [263](#). For instance, the skill "Gradient Descent" would have dependencies on "Partial Derivatives" and "Vector Calculus." This structure enables sophisticated adaptive sequencing. If a learner demonstrates weakness in a foundational skill, the system can automatically recommend remedial content before they attempt to master a more advanced one. This concept is supported by research into competency mapping, where skill trees are used to show dependencies between skills and couple them to conceptual ideas for more intuitive learning [132](#). Projects like `skill-it` demonstrate how dependency graphs can be constructed from data-driven analysis of tasks and skills associated with language models [337](#). Similarly, frameworks like the Dynamic Tech Radar and Competency Graph Model (DCGM) integrate a competency graph with other modules to create a dynamic representation of skills and their interconnections [162](#). This graph-based approach is the key to transforming the learning experience from a passive consumption of content to an active, guided journey of discovery.

To operationalize this architecture, a technology stack composed of open-source components is essential to ensure the system remains free and customizable. The foundation of the system would be an open-source Learning Management System (LMS)

or a custom-built platform. Options like Open edX, developed by Harvard and MIT, offer a robust, highly extensible platform capable of supporting self-paced, competency-based education [14](#). Its ability to be self-hosted provides complete control over data and governance policies [14](#). Another strong candidate is LearnHouse, an open-source platform designed specifically for creating and managing online courses with a focus on interactive elements like real-time code execution environments and auto-grading capabilities, which are crucial for validating technical competencies [69](#). Integrating with such an LMS would allow for the seamless delivery of content and management of user progress.

The second layer of the architecture is the Assessment & Analytics Layer, which acts as the engine driving the entire system. This layer's primary function is to collect data on learner performance and translate it into actionable insights. Data collection involves two streams: (1) results from formal assessments (quizzes, projects) and (2) logs of informal study activities. For the latter, a Learning Record Store (LRS) conforming to the Experience API (xAPI) standard is the ideal solution [68](#). xAPI allows for the granular tracking of any learning experience, whether online or offline. An open-source LRS like lxHive can be self-hosted, ensuring complete data privacy and control, a critical feature for a system built on free tools [31](#) [139](#). Simple scripts can be developed to capture interactions with various learning resources—videos watched on Khan Academy, articles read, code committed to a GitHub repository—and send them to the LRS as structured statements. This rich dataset becomes the raw material for the analytics engine.

The third layer, the User Interface & Experience, brings the system to life. It must be designed to be intuitive, motivating, and transparent. A central dashboard is non-negotiable, providing the learner with a clear visualization of their progress within the Skill Graph [36](#). This dashboard could display completed modules, current goals, learning velocity, and mastery levels for key skills [36](#) [329](#). Goal planning functionality is also essential, allowing users to set personal targets (e.g., "Master supervised learning in two months") and receive guidance on breaking them down into manageable steps [239](#). The ultimate goal of this layer is to make the adaptive nature of the system visible and understandable to the learner. When the system recommends a different path or a remedial resource, it should do so transparently, explaining the reasoning based on the analytics from the second layer. This transparency fosters trust and helps the learner take ownership of their educational journey. The integration of digital badges, issued upon the completion of a module or skill cluster, serves as a powerful motivator and provides a tangible record of achievement, leveraging open standards like Open Badges for portability and verification [27](#) [29](#). By combining these three layers—a skill-graph-based curriculum, an automated analytics engine, and a transparent user interface—it is

possible to construct a powerful, flexible, and truly personalized learning system that fully realizes the user's vision of replacing a rigid timeline with a fluid path to AI mastery.

Core Curriculum Design: Structuring Technical and Adjacent Knowledge Domains

The core curriculum of the proposed framework must be meticulously structured to cover both the deep technical foundations and the broader contextual domains essential for a well-rounded understanding of modern artificial intelligence. As specified by the user, this includes not only core technical skills but also adjacent fields such as AI ethics, systems design, and deployment infrastructure . The optimal approach to structuring this vast knowledge domain is not a simple linear progression but a hierarchical, dependency-aware model, best represented as a Skill Graph [129132](#). This graph-based curriculum will serve as the backbone for the framework's adaptive sequencing capabilities, ensuring that learners build a solid foundation before advancing to more complex topics. The curriculum can be logically divided into several major domains, each containing a series of interconnected modules and skills.

The first and most fundamental domain is **Mathematical Foundations**. Machine learning is inherently mathematical, and a solid grasp of its principles is non-negotiable for anyone aspiring to be an AI practitioner [61 63](#) . This domain must include Linear Algebra, Calculus, Probability, and Statistics [56 57](#) . Excellent, high-quality, and free resources exist for all these subjects. For beginners, Khan Academy provides a gentle introduction to concepts like linear algebra and calculus, making it an ideal starting point [185234](#). For those seeking a more rigorous, proof-based understanding, MIT OpenCourseWare offers comprehensive lecture series that are freely accessible [121122](#). Curated collections of books and guides, such as those found on GitHub and in PDF form, provide structured pathways for self-study, covering everything from vector calculus to matrix decompositions [94 96 149](#).

The second domain is **Programming and Software Engineering**. Proficiency in Python is the de facto standard for AI development [65](#) . For absolute beginners, Khan Academy's Computer Science program offers an excellent entry point into programming concepts without requiring prior experience [236](#). Beyond basic syntax, learners need to understand algorithm analysis, including time and space complexity, which is essential for optimizing machine learning models [237](#). The curriculum should also introduce version control with

Git, as it is a fundamental tool for any developer, and command-line interaction via the Bash terminal [65](#). The use of open-source libraries and frameworks is a critical part of the workflow. TensorFlow, for example, is a comprehensive, end-to-end open-source platform for machine learning that provides a rich ecosystem of tools and community resources [340](#). Other curated lists highlight numerous other open-source frameworks and libraries that are essential for building and deploying AI systems [370371](#).

The third and largest domain is **Machine Learning and Deep Learning Theory and Practice**. This is where the theoretical foundations are applied to solve real-world problems. The consensus among leading practitioners points towards a "build-first" pedagogical approach, where learners gain a deep understanding of how ML systems work by constructing them from the ground up. The centerpiece of this domain in the proposed framework should be the curriculum developed for Harvard University's CS249r course, *Designing Machine Learning Systems* [19](#). This book, combined with its associated TinyTorch project, offers a 20-module curriculum that teaches machine learning as computational systems engineering [20](#) [22](#). In TinyTorch, students implement PyTorch's core components—including tensors, automatic differentiation, optimizers, and neural networks—by building a library compatible with the PyTorch API [71](#) [72](#). This hands-on, bottom-up approach provides an unparalleled understanding of the inner workings of modern deep learning frameworks [75](#). Complementing this systems-level view, the fast.ai course, *Practical Deep Learning for Coders*, is repeatedly cited as a premier resource for applying deep learning to practical problems [48](#) [51](#). Its methodology demystifies complex topics by focusing on hands-on application with popular libraries like PyTorch, making it accessible to those with some coding experience [85](#) [153](#). Together, these two resources—one focused on the "why" and "how" from first principles, and the other on the "what" and "so what" of practical application—provide a complete and robust foundation in ML/DL.

The fourth domain covers **Adjacent and Contextual Knowledge**, which is crucial for developing responsible, deployable, and effective AI systems. This includes:

- **AI Ethics and Governance:** Understanding the societal impacts and ethical implications of AI is paramount. The University of Helsinki's free MOOC, "Ethics of AI," is consistently recommended as a top-tier resource for fostering ethical thinking [15](#) [16](#) [18](#). This course covers fairness, accountability, and transparency, providing a strong ethical grounding [18](#). To broaden this perspective, the curriculum can integrate materials from organizations like the OECD, UNESCO, and the EU, which provide frameworks and governance standards for responsible AI [361364365](#).
- **Machine Learning Systems and Deployment:** Building a model is only half the battle; deploying it reliably and scalably is the other. The Harvard CS249r textbook

and curriculum are again central here, covering the iterative process of designing ML systems that are deployable, reliable, and scalable [19](#) [23](#) . For deployment, the Cloud Native Computing Foundation (CNCF) provides essential training on cloud-native technologies, particularly Kubernetes, which is foundational for running modern AI workloads efficiently [9](#) [10](#) . CNCF offers numerous free training courses, such as "Introduction to Kubernetes" and "Kubernetes Fundamentals," which are invaluable for understanding the infrastructure side of AI [143144](#).

- **AI Literacy and Communication:** Beyond technical skills, professionals need AI literacy—the ability to understand, evaluate, and communicate about AI technologies [212214](#). Frameworks from the OECD outline the competencies needed, including understanding core concepts like algorithms, data, and models, as well as developing critical thinking and creativity [214361](#). Resources like Google's Professional Certificate in AI, while paid, demonstrate the value of hands-on practice with tools like Gemini and NotebookLM, which can be supplemented with free alternatives [43](#) [175](#).

The following table summarizes key free resources for each major domain, providing a concrete starting point for the curriculum.

Domain	Key Skills / Topics	Recommended Free Resources
Mathematical Foundations	Linear Algebra, Calculus, Probability, Statistics	Khan Academy 185 , MIT OpenCourseWare 121 , Mathematics for Machine Learning (Book) 62 , Various PDF Guides 94
Programming & Tools	Python Programming, Algorithm Analysis, Git, Bash	Khan Academy CS Program 236 , W3Schools Python Tutorial 65 , nStudy Platform Documentation 11
ML/DL Theory & Practice	Supervised/Unsupervised Learning, Neural Networks, NLP, CV	<i>Designing Machine Learning Systems</i> (Book) 19 , Harvard CS249r TinyTorch Curriculum 20 , fast.ai Practical DL for Coders Course 48
AI Ethics & Governance	Fairness, Accountability, Transparency, Societal Impact	University of Helsinki MOOC: Ethics of AI 15 , OECD AI Literacy Framework 361 , UNESCO/EU Governance Standards 364
ML Systems & Deployment	Scalability, Reliability, Infrastructure, CI/CD	Harvard CS249r Textbook 19 , CNCF Training Courses (Kubernetes) 143 , Linux Foundation Materials 10
AI Literacy & Communication	Understanding AI Concepts, Prompt Engineering, Critical Thinking	Google AI Professional Certificate Activities 175 , Various Online Guides 44

By organizing these diverse but interconnected topics into a skill graph, the framework ensures that learners follow a logical progression. For example, the node for "Linear Regression" would depend on nodes for "Matrix Multiplication" and "Gradient Descent." The node for "Building a Recommendation System" would depend on nodes for "Collaborative Filtering," "Data Pipelines," and "Model Evaluation." This dependency-aware structure is the key to creating a truly adaptive learning path that meets the

learner at their level and guides them forward based on their demonstrated mastery, fulfilling the core requirement of a flexible, trackable system.

Implementing Competency Assessment: Tools for Demonstrating Mastery

A cornerstone of the proposed competency-based framework is the implementation of robust assessment mechanisms that move beyond simple completion tracking to measure genuine understanding and skill proficiency . The user explicitly requested a mix of self-assessment tools and automated insights, which together form a comprehensive evaluation system . This section outlines the strategies for defining mastery, selecting appropriate assessment tools, and automating the grading and feedback process. The overarching goal is to create a system where advancement to the next module is contingent upon demonstrated mastery of the preceding one, a core tenet of mastery learning [34](#) [66](#) .

Defining "mastery" is the first step. It implies a high degree of expertise and proficiency, distinguishing it from mere competence [34](#) . In a practical sense, mastery can be operationalized through a combination of formative and summative assessments. Formative assessments are low-stakes checks for understanding integrated throughout the learning process. These include quizzes, interactive exercises, and short coding challenges. The fast.ai course, for example, is built around interactive notebooks and embedded quizzes that allow learners to immediately test their understanding of a concept [88](#) . Summative assessments are higher-stakes evaluations that occur at the end of a module or a significant learning segment. These typically take the form of larger projects that require the integration of multiple skills. The TinyTorch curriculum itself serves as a massive summative assessment, requiring students to build a functional deep learning framework from scratch, thereby demonstrating mastery of tensors, autograd, optimization, and neural network architectures [71](#) [73](#) .

For self-assessment, a variety of tools can be integrated. Platforms like Quizizz offer a free tier that allows educators and learners to generate AI-powered quiz questions from any topic or even uploaded documents like PDFs, providing a quick and easy way to test knowledge [31](#) . For coding-related skills, platforms like LearnHouse provide built-in playgrounds with auto-grading capabilities for code submissions in over 30 programming languages, directly validating a learner's ability to execute a task correctly [69](#) . Project

rubrics are another critical component of self-assessment, especially for creative or open-ended tasks. These rubrics should clearly define the criteria for success, allowing the learner to evaluate their own work against established benchmarks. The development of scenario-based self-assessment instruments, such as AICompAss, provides a model for creating structured evaluations of complex competencies across multiple domains [163](#).

Automated grading and feedback are crucial for scaling this system for a self-directed learner. For objective assessments like multiple-choice quizzes or code that can be checked against predefined outputs, automation is straightforward. More challenging is the automated assessment of open-response questions, essays, or complex projects. While this is an area of active research, AI-powered grading engines are becoming increasingly sophisticated [293](#). For example, MagicSchool AI offers a free tier that can generate rubrics and provide feedback on generated content [31](#). The EdOptimize platform, an open-source project, analyzes assessment data to provide detailed item-level performance metrics, which could be adapted to provide learners with insights into their specific strengths and weaknesses [13](#).

To manage the validation of competencies, the system can employ different models for determining mastery. One common heuristic is to require a certain threshold of correctness, such as answering a set number of questions consecutively or achieving a minimum score on a quiz [238](#). This is a simple yet effective rule-based approach. A more sophisticated method is Bayesian Knowledge Tracing (BKT), a family of probabilistic models widely used in intelligent tutoring systems to estimate a student's knowledge state and predict their likelihood of success on future problems [238](#). While BKT requires more complex implementation, simpler Elo-rating-like systems, inspired by their use in chess, have proven effective. Khan Academy uses such a system to update a learner's mastery level for a specific skill based on their performance on related quiz questions; getting a question right against a lower-rated opponent increases your rating more than getting it right against a higher-rated one [313346](#). This provides a dynamic and nuanced measure of proficiency that goes beyond simple pass/fail.

The following table compares different assessment methods and tools suitable for integration into the framework.

Assessment Type	Description	Example Tools & Methods	Strengths	Weaknesses
Quizzes & MCQs	Low-stakes tests of factual knowledge and basic concepts.	Quizizz (AI-generated), Custom quizzes linked to content, Fast.ai interactive notebooks 88	Automated grading, immediate feedback, good for formative checks.	Limited ability to assess deep understanding or complex problem-solving.
Coding Exercises	Practical tasks requiring the implementation of code to solve a specific problem.	LearnHouse Code Playground 69 , WakaTime (for logging activity) 1 , HazyResearch/skill-it 337	Directly assesses programming skills, automated validation of output.	Can be difficult to grade for style, efficiency, or correctness of logic.
Projects	Larger, open-ended tasks that require the integration of multiple skills.	TinyTorch curriculum 71 , Kaggle competitions, custom capstone projects.	Assesses deep understanding, problem-solving, and the ability to apply knowledge in novel contexts.	Time-consuming to create, grade, and provide meaningful feedback on.
Auto-Graded Labs	Interactive environments where learners write code and receive immediate feedback on correctness.	Google Cloud AI classes (interactive labs) 176 , Various Jupyter notebook environments.	Combines practical application with instant feedback, simulates a real development environment.	Requires a dedicated backend environment to run securely.
Rubric-Based Evaluation	Self-assessment using a predefined scoring guide that breaks down a complex task into criteria.	Custom rubrics created in Google Docs, Moodle's built-in rubric tool, MagicSchool AI 31 .	Allows for subjective evaluation of quality, creativity, and presentation.	Subjectivity can lead to inconsistent self-assessment; relies on learner honesty.

Finally, to provide tangible recognition of achievement, the framework should issue credentials upon completing modules or clusters of skills. Digital badges, governed by the Open Badges specification, are an ideal solution [27](#). They are portable, verifiable, and can contain metadata describing the skill they represent and the criteria for earning it [29](#). This creates a public-facing portfolio of accomplishments that a learner can share on a resume or LinkedIn profile, adding significant value to the self-directed learning process [83](#). Issuing badges for completing foundational math modules, finishing the TinyTorch curriculum, or passing a capstone project would provide clear milestones and a powerful sense of accomplishment, reinforcing the learner's motivation throughout their journey.

Automating Progress Analytics: The Engine of Personalized Feedback

The heart of a truly adaptive and personalized learning framework lies in its ability to analyze learner data and provide timely, actionable feedback. This capability transforms the system from a static repository of information into a responsive learning companion. The proposed framework aims to achieve this by combining manual self-assessment data

with automatically logged study activity, feeding this information into an analytics engine that can deliver descriptive, diagnostic, predictive, and prescriptive insights . This section details the architecture for collecting this data, the types of analytics that can be performed, and the tools available to implement them using a free and open-source stack.

The foundation of the analytics engine is a robust data collection mechanism. As previously discussed, this involves capturing two distinct streams of data: assessment results and activity logs. For activity data, adopting the Experience API (xAPI) standard is the most effective approach. xAPI is designed to be highly flexible, allowing any learning experience to be tracked and recorded as a simple statement of the form "Actor verb Object" (e.g., "learner_123 completed module_calculus_1"). An open-source Learning Record Store (LRS) like lxHive can be self-hosted to store these statements securely [139](#). To populate the LRS, lightweight scripts can be deployed to monitor a learner's interactions with various resources. For example, a script could listen for events when a video on Khan Academy finishes playing, when a learner commits code to a GitHub repository, or when a notebook in a Jupyter environment is saved. These scripts would then package the relevant information into an xAPI statement and send it to the LRS. This creates a comprehensive longitudinal profile of the learner's engagement with the curriculum [68](#) .

With this rich dataset in place, the system can perform four core types of learning analytics, as defined by the Society for Learning Analytics Research [12](#) : 1. **Descriptive Analytics:** This answers the question "What happened?" by summarizing past events. For a learner, this could manifest as a dashboard showing completion rates for modules, total time spent studying, or a breakdown of time spent per subject. Open-source dashboards like those provided by the EdOptimize project can visualize curriculum pacing and skill performance over time, offering a high-level overview of progress [13](#) . 2. **Diagnostic Analytics:** This answers "Why did it happen?" by identifying root causes. If a learner struggles with a particular concept, diagnostic analytics can help pinpoint the reason. For example, if a learner fails a quiz on backpropagation, the system could analyze their activity logs to see if they skipped the prerequisite videos on matrix calculus or spent very little time on the interactive coding exercises. This type of analysis helps differentiate between a lack of conceptual understanding versus a failure to engage with the material [12](#) . 3. **Predictive Analytics:** This uses historical data to forecast future outcomes. By analyzing patterns in a learner's past performance, the system can predict their likelihood of succeeding in an upcoming module or failing a specific assessment. Machine learning models can be trained on aggregated data from many learners to identify risk factors associated with poor performance, such as low quiz scores on foundational topics or infrequent engagement [12](#) [255](#). This allows the system to proactively intervene before a

learner falls too far behind. 4. **Prescriptive Analytics:** This answers "What should be done?" by recommending specific actions. Based on the insights from the other three types of analytics, the system can suggest targeted interventions. For example, if predictive analytics flags a learner as being at risk for failing a module on neural networks, prescriptive analytics might recommend a specific remedial article, a simpler coding exercise, or a review of a particular video lecture [12](#). This is the highest level of adaptivity and is the ultimate goal of the analytics engine.

The implementation of these analytics can range from simple to highly sophisticated. At a basic level, the system can use rule-based logic derived from the Skill Graph. For instance, if a learner scores below 80% on a pre-test for Module B, and Module A is a strict prerequisite, the system can automatically block access to Module B and redirect the learner to Module A's content. This is a form of adaptive sequencing that is relatively straightforward to implement and is used in platforms like Canvas and Moodle [378](#). A more advanced implementation would involve building models that track a learner's mastery state over time. The Bayesian Knowledge Tracing (BKT) model, mentioned earlier, is a prime example of such a technique, estimating the probability that a learner knows a skill after each interaction [238](#). While implementing BKT from scratch is complex, its principles can inspire simpler heuristics.

Several open-source platforms and tools can accelerate the development of this analytics layer. The Studyield platform, for instance, provides a detailed example of an open-source AI-powered learning platform with a dedicated 'Progress Analytics' module that monitors learning velocity and mastery levels [36](#). The EdOptimize platform offers a suite of open-source R-based dashboards for analyzing curriculum analytics, platform usage, and implementation effectiveness, which could be adapted for individual learner analysis [13](#). Furthermore, open-source LMS platforms like Moodle and Open edX come with their own built-in analytics suites and APIs that can be leveraged or extended [12](#) [14](#). For example, Open edX's RESTful APIs allow for integration with external analytics services, enabling the creation of custom dashboards and reports [379](#). The key is to start with descriptive and diagnostic analytics, which provide clear value by giving the learner visibility into their own behavior, and then progressively layer on predictive and prescriptive capabilities as the system matures. This data-driven approach ensures that the framework is not just a collection of links but a genuinely intelligent and responsive learning environment.

Operationalizing the Framework: Technology Stack and Implementation Roadmap

Translating the architectural and curricular designs into a functional, end-to-end learning system requires a carefully selected technology stack and a phased implementation plan. The goal is to create a self-contained, customizable, and privacy-preserving platform that leverages the best of the free and open-source software ecosystem. This final section outlines a recommended technology stack and a pragmatic roadmap for building and deploying the modular, competency-based AI learning framework.

The core of the system should be a self-hosted Learning Management System (LMS) or a custom-built platform that prioritizes extensibility. Two strong candidates emerge from the research: 1. **Open edX**: Developed by Harvard and MIT, Open edX is a powerful, free, and open-source platform used by major institutions worldwide [14](#). Its key advantages are its maturity, scalability, and extensive customization options through XBlocks and APIs [14 379](#). It already supports features like self-paced learning, robust analytics dashboards, and the issuance of digital credentials, making it an excellent foundation [14 379](#). Its integration with the Learning Tools Interoperability (LTI) standard would also facilitate the embedding of external adaptive learning tools, like Harvard's 'Bridge for Adaptivity' [377](#). 2. **LearnHouse**: This is a newer, more modern alternative that appears purpose-built for the kind of interactive, competency-based learning envisioned here. It features a block-based editor similar to Notion, real-time code playgrounds with auto-grading, and a focus on creating engaging course content [69](#). Its AGPL-3.0 license and ease of self-hosting via a CLI make it a compelling choice for rapid development [69](#).

For the analytics backbone, integrating an open-source Learning Record Store (LRS) conforming to the xAPI standard is essential for capturing detailed activity data. **lxHive** is a lightweight and fast open-source LRS that can be easily self-hosted, ensuring that all learner data remains private and under the user's control [31 139](#). This data will feed into the analytics engine.

The front-end user interface can be developed using modern web technologies like React or Vue.js, allowing for the creation of a dynamic and responsive dashboard. This dashboard should visually represent the Skill Graph, showing the learner's current position, completed nodes, and recommended next steps. It should also feature the analytics dashboards described previously, providing insights into study habits and performance trends [13 36](#).

The following table summarizes the recommended technology stack for the framework:

Component	Recommended Technology	Justification
Core Platform	Open edX 14 or LearnHouse 69	Both are open-source, self-hostable, and designed for creating and managing online courses with a focus on customization and interactive content.
Activity Tracking	lxHive 139 + xAPI Statements	Provides a standardized, flexible, and privacy-preserving method for logging all learning activities. lxHive is a lightweight, open-source LRS.
Analytics Engine	Custom Backend (e.g., Python/FastAPI) 69 + R 13 or JavaScript	A custom backend can process data from the LRS and implement rules or models for mastery assessment. R can be used for statistical analysis and generating reports.
User Interface	React/Vue.js	Modern JavaScript frameworks for building dynamic, single-page applications with a responsive and interactive user experience.
Database	PostgreSQL 36	A robust, open-source relational database for storing user accounts, course data, and other structured information.
Version Control	Git / GitHub	Essential for managing the curriculum content, platform code, and learner projects.

With the technology stack defined, a phased implementation roadmap can ensure steady progress and a functional minimum viable product (MVP). This roadmap consists of four phases:

Phase 1: Minimum Viable Product (MVP) - Foundational Curriculum and Basic

Tracking The initial goal is to create a small, self-contained learning path and establish the basic mechanics of the system. This phase would involve:

- Selecting a narrow but critical segment of the curriculum, such as the mathematical prerequisites for machine learning (Linear Algebra, Calculus).
- Curating a set of high-quality free resources for this segment and integrating them into the chosen LMS (e.g., linking to Khan Academy videos and MIT OCW readings).
- Creating a simple assessment system, perhaps by integrating with a third-party quiz tool or manually creating quizzes within the LMS.
- Developing a basic progress-tracking dashboard that simply shows which modules a user has "completed" (i.e., passed an assessment for). The assessment logic would be a simple pass/fail based on a minimum score. The goal is to replicate the core concept of mastery learning—proceed only upon demonstration of proficiency—on a small scale.

Phase 2: Automation and Diagnostic Analytics Once the MVP is functional, the focus shifts to automating data collection and providing deeper insights.

- Integrate the xAPI-compliant LRS (e.g., lxHive) and develop scripts to automatically log key study activities (e.g., video playback completion, time spent on a page).
- Enhance the user dashboard to include diagnostic analytics. For example, show a chart of daily study time, or flag modules where the user's quiz scores were low compared to their overall average.
- Implement a more sophisticated mastery model, such as an Elo-rating-like system, to provide a continuous measure of proficiency for each skill rather than a binary pass/fail status [346](#).

Phase 3: Adaptive Sequencing and Prescriptive Feedback This phase introduces the "adaptive" element of the framework, moving beyond reactive tracking to proactive guidance.

- Develop a rule-based adaptive engine. For example, if a user scores below a certain threshold on a post-module quiz, the system automatically presents them with a "remediation path"—a set of simpler exercises or review materials related to the weak concepts—before allowing them to retake the quiz.
- Begin populating the Skill Graph by formally mapping out the dependencies between the modules in the MVP. This graph will become the decision-making logic for the adaptive engine.
- Implement a basic badge-issuance system using the Open Badges standard to provide tangible rewards for completing modules [27](#).

Phase 4: Expansion and Advanced Features With a working adaptive core, the framework can be expanded to cover the entire curriculum and incorporate more advanced AI-powered features.

- Gradually add new modules to the curriculum, always ensuring they are properly mapped within the Skill Graph with their dependencies defined.
- Explore more advanced mastery modeling techniques, potentially moving from heuristics to implementations of Bayesian Knowledge Tracing (BKT) to better predict a learner's knowledge state [238](#).
- Integrate generative AI agents for enhanced tutoring. Inspired by systems like Khanmigo and Studyield's Teach-Back feature, these agents could provide personalized explanations, generate practice problems, or even critique a learner's code, all grounded in the specific content of the curriculum [36](#) [191](#).

By following this structured roadmap, it is entirely feasible to build a powerful, flexible, and personalized AI learning framework from the ground up using only free and open-source tools. This approach not only fulfills the user's specific requirements but also creates a sustainable and extensible system for lifelong learning in the rapidly evolving field of artificial intelligence.

Reference

1. Time tracking tools for your coding tasks - Livable Software <https://livablesoftware.com/time-tracking-coding-tools/>
2. Cloud Native Computing Foundation (CNCf) <https://www.cncf.io/>
3. CNCf Cloud Native AI White Paper <https://tag-runtime.cncf.io/wgs/cnaiwg/whitepapers/cloudnativeai/>
4. Cloud Native Developer IT Professional Program - Linux Foundation <https://training.linuxfoundation.org/training/cloud-native-developer-it-professional-program/>
5. Cloud Native Training Courses | CNCf <https://www.cncf.io/training/courses/>
6. Certified Cloud Native Platform Engineering Associate (CNPA) <https://training.linuxfoundation.org/certification/certified-cloud-native-platform-engineering-associate-cnpa/>
7. Generative AI development is going cloud native and open source ... <https://lfaidata.foundation/blog/2024/04/23/generative-ai-development-is-going-cloud-native-and-open-source-how-to-make-sense-of-it-and-whats-next/>
8. Page 2 – Cloud Native Computing Foundation <https://www.cncf.io/u/0/page/2/>
9. Cloud Native AI: Opportunities and Insights | PDF - Scribd <https://www.scribd.com/document/811284432/CNCf-AI-3>
10. Contributed Archives - Page 2 of 10 - Linux Foundation - Education <https://training.linuxfoundation.org/blog/category/contributed/page/2/>
11. nStudy: Software for Learning Analytics about Processes for Self ... <https://learning-analytics.info/index.php/JLA/article/view/6194>
12. Top 10 Learning Analytics Tools: Overview, Types & Examples <https://www.coschool.ai/feeds/blog/learning-analytics-tools>
13. EdOptimize - An Open Source K-12 Learning Analytics Platform <https://github.com/PlaypowerLabs/EdOptimize>

14. Open edX: Home <https://openedx.org/>
15. Ethics of AI - MOOC.fi <https://ethics-of-ai.mooc.fi/>
16. MOOC: AI in Society | Una Europa <https://www.una-europa.eu/study/mooc-ai-society>
17. UNESCO and LG AI to Develop a Global Massive Open Online ... <https://www.unesco.org/en/articles/unesco-and-lg-ai-develop-global-massive-open-online-course-ethics-artificial-intelligence>
18. The Learning Curve Is the Moat. Courses and Certificate to enable... <https://medium.com/@adnanmasood/the-learning-curve-is-the-moat-d94178539f0d>
19. [PDF] Designing Machine Learning Systems <https://18636251.s21i.faiusr.com/61/ABUIABA9GAAghIK0ugYowM2h3QY.pdf>
20. Tiny Torch Building Machine Learning Systems from First Principles <https://arxiv.org/html/2601.19107v2>
21. Machine Learning Systems: Design and Implementation <https://openmlsys.github.io/html-en/>
22. Machine Learning Systems <https://mlsysbook.ai/>
23. Design a machine learning system - Chip Huyen <https://huyenchip.com/machine-learning-systems-design/design-a-machine-learning-system.html>
24. Machine Learning Engineering: Complete Guide to Building ... <https://www.databricks.com/blog/machine-learning-engineering-complete-guide-building-production-ml-systems>
25. Practical Machine Learning System Design - Leanpub <https://leanpub.com/practical-machine-learning-system-design>
26. ml-system-design.md - Machine-Learning-Interviews - GitHub <https://github.com/alirezadir/machine-learning-interviews/blob/main/src/MLSD/ml-system-design.md>
27. Open Badges - 1EdTech <https://www.1edtech.org/standards/open-badges>
28. Badges API - Moodle Developer Resources <https://moodledev.io/docs/5.2/apis/subsystems/badges>
29. Open Badges 2.0 & 3.0 Guide | IssueBadge <https://issuebadge.com/guides/open-badges-standard>
30. About Open Badges <https://openbadgefactory.com/en/about-open-badges/>
31. Open-Source AI Education Tools — What's Available for Free <https://www.edugenius.app/blog/open-source-ai-education-tools-free>
32. ai-education · GitHub Topics <https://github.com/topics/ai-education>
33. Can Mastery-Based Learning Replace Seat Time? <https://www.aei.org/op-eds/can-mastery-based-learning-replace-seat-time/>

34. What is Mastery-Based Learning? - Modern Classrooms Project <https://www.modernclassrooms.org/blog/what-is-mastery-based-learning>
35. A Practical Review of Mastery Learning - PMC <https://pmc.ncbi.nlm.nih.gov/articles/PMC10159400/>
36. Studyield - GitHub <https://github.com/studyield/studyield>
37. How to Learn AI in 2026? The Complete Roadmap <https://ischool.syracuse.edu/how-to-learn-ai/>
38. How to Learn AI in 2025: The Essential Skills Roadmap for Beginners <https://blog.dataengineerthings.org/how-to-learn-ai-in-2025-the-essential-skills-roadmap-for-beginners-4473268012a9>
39. (PDF) Roadmap to Empower Researchers in AI Skills in Developing ... https://www.researchgate.net/publication/389389797_Roadmap_to_Empower_Researchers_in_AI_Skills_in_Developing_Countries
40. AI Learning Roadmap: From Beginner to Expert (2026) - Coursera <https://www.coursera.org/resources/ai-learning-roadmap>
41. Complete Detailed Roadmap To Learn AI In 2025 With ... - YouTube <https://www.youtube.com/watch?v=s3KnSb9b4Pk>
42. Free AI Learning Tools: Top Resources for Self-Directed Learning <https://www.leveragai.com/free-ai-learning-tools-top-resources-for-self-directed-learning>
43. Understanding AI: AI tools, training, and skills - Google AI <https://ai.google/learn-ai-skills/>
44. 5 Essential AI Skills to Master in 2025 (Complete Learning Guide) <https://www.techaffiliate.in/blog/ai-skills-2025-complete-career-guide>
45. Top In Demand AI Skills (2025) - Skillsoft <https://www.skillsoft.com/blog/essential-ai-skills-everyone-should-have>
46. How to Learn AI From Scratch in 2026: A Complete Expert Guide <https://www.datacamp.com/blog/how-to-learn-ai>
47. fast.ai—Making neural nets uncool again – fast.ai <https://www.fast.ai/>
48. Practical Deep Learning for Coders - Fast.ai <https://course.fast.ai/>
49. AI Courses - DeepLearning.AI <https://www.deeplearning.ai/courses>
50. Free online machine learning curriculum - Chip Huyen <https://huyenchip.com/2019/08/05/free-online-machine-learning-curriculum.html>
51. fast.ai releases new deep learning course, four libraries, and 600 ... <https://www.fast.ai/posts/2020-08-21-fastai2-launch.html>

52. 20+ Best Free Machine Learning Courses - DataTalks.Club <https://datatalks.club/blog/free-machine-learning-courses.html>
53. offchan42/machine-learning-curriculum: :computer - GitHub <https://github.com/offchan42/machine-learning-curriculum>
54. AI Machine Learning Roadmap: Self Study AI! - YouTube <https://www.youtube.com/watch?v=nznFtfgP2ks>
55. Best AI and Machine Learning Courses in 2025 https://learnprompting.org/blog/ai-and-machine-learning-courses?srsltid=AfmBOorF9LBULb_IRLYuH-LXpnNCIHyinYuKMLyqnwFUaBdkqu-wl93-
56. Mathematics for Machine Learning and Data Science <https://www.deeplearning.ai/specializations/mathematics-for-machine-learning-and-data-science>
57. Math for Machine Learning: Linear Algebra, Calculus, and ... <https://www.amazon.com/Math-Machine-Learning-Probability-Explained-ebook/dp/B0FT4T5ZYJ>
58. How To Learn Math for Machine Learning FAST (Even With Zero ... <https://www.youtube.com/watch?v=KgolhE7p-KY>
59. Master the essential Mathematics for Machine Learning - Facebook <https://www.facebook.com/classcentral/posts/master-the-essential-mathematics-for-machine-learning-linear-algebra-calculus-pr/1173892398111090/>
60. Should I learn math first (such as Linear Algebra, Calculus ... - Quora <https://www.quora.com/Should-I-learn-math-first-such-as-Linear-Algebra-Calculus-and-Probability-before-starting-AI-ML-courses-or-is-the-math-usually-taught-as-part-of-the-AI-ML-courses-themselves-so-I-dont-need-to-study-the-theory>
61. Maths for Machine Learning - GeeksforGeeks <https://www.geeksforgeeks.org/machine-learning/machine-learning-mathematics/>
62. Why Linear Algebra, Calculus, and Probability for ML - LinkedIn https://www.linkedin.com/posts/anil-ananthaswamy_the-theoretical-minimum-for-machine-learning-activity-7310167102378688514-1lGQ
63. Calculus, Math and Linear Algebra for Machine Learning - Medium <https://medium.com/@koushikkushal95/building-the-foundation-calculus-math-and-linear-algebra-for-machine-learning-91a95e2f36b7>
64. Linear Algebra, Probability, Calculus, Optimization [Lecture 1] <https://www.youtube.com/watch?v=C8hEa2qb46k>
65. Prerequisites and prework | Machine Learning | Google for Developers <https://developers.google.com/machine-learning/crash-course/prereqs-and-prework>
66. Mastery learning in CS1: a longitudinal study during and post ... <https://link.springer.com/article/10.1007/s44217-024-00361-x>

67. Modularization for mastery learning in CS1: a 4-year action research ... <https://pmc.ncbi.nlm.nih.gov/articles/PMC10018628/>
68. XAPI - THE CASE STUDIES | Uhlberg Advisory https://uhlberg-advisory.de/wp-content/uploads/2017/06/HT2_-_Case-Studies-xAPI.pdf
69. The Next-gen Open Source learning platform for everyone · GitHub <https://github.com/learnhouse/learnhouse>
70. harvard-edge/cs249r_book: Machine Learning Systems - GitHub https://github.com/harvard-edge/cs249r_book
71. Tiny Torch Building Machine Learning Systems from First Principles <https://arxiv.org/html/2601.19107v1>
72. cs249r_book/tinytorch/README.md at dev - GitHub https://github.com/harvard-edge/cs249r_book/blob/dev/tinytorch/README.md
73. [PDF] TinyTorch: Building Machine Learning Systems from First Principles https://mlsysbook.ai/tinytorch/_static/downloads/TinyTorch-Paper.pdf
74. Introduction to Machine Learning Systems - GitHub https://github.com/vanHeemstraSystems/cs249r_book-machine-learning
75. Harvard just open-sourced an entire ML Systems curriculum. - Threads <https://www.threads.com/@datasciencereality/post/DTH-UEfk4p0/harvard-just-open-sourced-an-entire-ml-systems-curriculum-the-complete-learning>
76. Harvard Open-Sources AI Engineering Curriculum | Ultan O. posted ... https://www.linkedin.com/posts/ultanorourke_harvard-just-open-sourced-an-entire-ml-systems-activity-7413991930705104898-_xEn
77. Harvard released a Free Book on ML system engineering - Facebook <https://www.facebook.com/groups/techtitansgroup/posts/1539165607410752/>
78. Build Machine Learning Systems with TinyTorch | PDF - Scribd <https://www.scribd.com/document/980951708/TinyTorch-Paper>
79. Ethics of AI, MOOC 1/9/2024–31/8/2025 | Courses | Studies Service <https://studies.helsinki.fi/courses/course-implementation/otm-e6fbf97e-5bba-495a-9d3f-e7bbc4a36e89>
80. Free Course: Ethics of AI from University of Helsinki | Class Central <https://www.classcentral.com/course/helsinki-ethics-of-ai-98596>
81. University of Helsinki offers online course Ethics of AI in Finnish and ... <https://www.eurekalert.org/news-releases/935681>
82. Online Training 'Ethics of AI', Helsinki | Cities for Digital Rights <https://citiesfordigitalrights.org/online-training-ethics-ai-helsinki>

83. University of Helsinki are offering free course in AI. After finishing ... https://www.reddit.com/r/learnprogramming/comments/c7lkf7/university_of_helsinki_are_offering_free_course/
84. Studying virtually at the University of Helsinki <https://www.unaeuropastudies.uzh.ch/en/dsai-me/dsai-me-uh.html>
85. Free Course: Practical Deep Learning For Coders from fast.ai <https://www.classcentral.com/course/independent-practical-deep-learning-for-coders-7887>
86. Practical Deep Learning for Coders - GitHub <https://github.com/fastai/course22-web>
87. Deep Learning For Coders—36 hours of lessons for free - Fast.ai <http://course18.fast.ai/>
88. Practical Deep Learning for Coders: Lesson 1 - YouTube https://www.youtube.com/watch?v=8SF_h3xF3cE
89. Practical Deep Learning for Coders, Chapter 1 - Kaggle <https://www.kaggle.com/code/jhoward/practical-deep-learning-for-coders-chapter-1>
90. The book - Practical Deep Learning for Coders <https://course.fast.ai/Resources/book.html>
91. Practical Deep Learning for Coders: Lesson 1 - Daniel Roy Greenfeld <https://daniel.feldroy.com/posts/course-fastai-lesson-01>
92. Practical Deep Learning for Coders - Part 2 overview <https://course.fast.ai/Lessons/part2.html>
93. [PDF] Marc Peter Deisenroth A. Aldo Faisal Cheng Soon Ong <https://mml-book.github.io/book/mml-book.pdf>
94. Mathematics for Machine Learning Guide | PDF - Scribd <https://www.scribd.com/document/932243418/MATHEMATICS-FOR-MACHINE-LEARNING>
95. [PDF] Mathematics for Machine Learning <https://course.ccs.neu.edu/ds4420sp20/readings/mml-book.pdf>
96. rohanmistry231/Mathematics-for-Machine-Learning-Books - GitHub <https://github.com/rohanmistry231/Mathematics-for-Machine-Learning-Books>
97. Mathematics for Machine Learning - DOKUMEN.PUB <https://dokumen.pub/mathematics-for-machine-learning-q-3594475.html>
98. Mathematics of Machine Learning: Master linear algebra, calculus ... <https://www.clcoding.com/2025/05/mathematics-of-machine-learning-master.html>
99. [PDF] Mathematical Analysis of Machine Learning Algorithms - Tong Zhang <https://www.tongzhang-ml.org/lt-book/lt-book.pdf>
100. Mathematics For Machine Learning Full Course - Part 1 - YouTube <https://www.youtube.com/watch?v=wPJBvmV70Y8>

101. [PDF] Mathematical Foundations of Machine Learning - Seongjai Kim https://skim.math.msstate.edu/LectureNotes/Machine_Learning_Lecture.pdf
102. The Future of Education: A Systematic Literature Review of Self ... https://www.researchgate.net/publication/394570813_The_Future_of_Education_A_Systematic_Literature_Review_of_Self-Directed_Learning_with_AI
103. Essential AI Skills For 2026 - YouTube <https://www.youtube.com/watch?v=sHDCgAWWpNQ>
104. Understanding Parents' Perspectives on Responsible AI for ... <https://dl.acm.org/doi/10.1145/3772318.3790479>
105. [PDF] Leveraging Generative AI in Support of Self-Directed Learning https://www.e-iji.net/dosyalar/iji_2026_2_15.pdf
106. Top AI Skills of 2025 That Will Define 2026 and Beyond <https://www.lifelonglearningsg.org/resources/articles/top-ai-skills>
107. [PDF] Best Practices in AIED - World Digital Education Alliance <https://wdea.bnu.edu.cn/docs/2026-05/8ae1cea3c3654e558c3f485ec361155c.pdf>
108. Learner agency in self-directed use of GenAI for informal language ... <https://www.tandfonline.com/doi/full/10.1080/17501229.2026.2648781>
109. [PDF] OECD Digital Education Outlook 2026 (EN) https://www.oecd.org/content/dam/oecd/en/publications/reports/2026/01/oecd-digital-education-outlook-2026_940e0dd8/062a7394-en.pdf
110. Top 5 AI Skills to Learn in 2026 for Future Careers | UniAthena <https://uniathena.com/top-ai-skills-to-learn>
111. [PDF] A Memetic Walrus Algorithm with Expert-guided Strategy for ... - arXiv <https://arxiv.org/pdf/2506.13092>
112. Adaptive learning with human factors and Artificial Intelligence <https://www.sciencedirect.com/science/article/pii/S0169814125001404>
113. Revolutionizing biotech and pharmaceutical education with adaptive ... <https://www.frontiersin.org/journals/education/articles/10.3389/feduc.2025.1679222/full>
114. AI for All: Adaptive, Accessible, and Inclusive Learning Experiences ... <https://www.mdpi.com/2078-2489/17/2/216>
115. [PDF] Curriculum Group Policy Optimization: Adaptive Sampling for ... https://openaccess.thecvf.com/content/CVPR2026/papers/Li_Curriculum_Group_Policy_Optimization_Adaptive_Sampling_for_Unleashing_the_Potential_CVPR_2026_paper.pdf
116. A deep learning-based intelligent curriculum system for enhancing ... <https://www.nature.com/articles/s41598-025-27153-z>

117. [PDF] Adaptive Test Recommendation for Mastery Learning - LRE <https://www.lre.epita.fr/dload/papers/bouarour.23.dse.pdf>
118. Day 21: Learning Path Generator - Intelligent Curriculum Sequencing <https://aieworks.substack.com/p/day-21-learning-path-generator-intelligent>
119. Adaptive learning - Grokipedia https://grokipedia.com/page/Adaptive_learning
120. [PDF] Self-Paced Curriculum Learning - Semantic Scholar <https://www.semanticscholar.org/paper/Self-Paced-Curriculum-Learning-Jiang-Meng/21d255246cd7ddba24a651fd716950f893ea8eb2>
121. MIT OpenCourseWare | Free Online Course Materials <https://ocw.mit.edu/>
122. 7 free online MIT courses to grasp machine learning - Medium <https://medium.com/open-learning/7-free-online-mit-courses-to-grasp-machine-learning-3ef5d819bbd8>
123. Explore the world of artificial intelligence with online courses from MIT <https://openlearning.mit.edu/news/explore-world-artificial-intelligence-online-courses-mit>
124. Free AI courses from MIT - Facebook <https://www.facebook.com/groups/aiworldopen/posts/1342254064375814/>
125. 13 Foundational AI Courses, Resources from MIT <https://www.ocw-openmatters.org/2025/05/21/13-foundational-ai-courses-resources-from-mit/>
126. I Have Compiled the Top 9 - Free - AI courses. | Alex Barády - LinkedIn https://www.linkedin.com/posts/alexbarady_i-have-compiled-the-top-9-free-ai-courses-activity-7391806540854243328-Tib_
127. Lec 01. Introduction to Deep Learning - MIT OpenCourseWare https://ocw.mit.edu/courses/6-7960-deep-learning-fall-2024/resources/mit6_7960f24_lec01_mp4/
128. MIT Introduction to Deep Learning (2025) | 6.S191 - YouTube <https://www.youtube.com/watch?v=alfdI7S6wCY>
129. Modeling Competency Dependencies with Graph AI | iCAN <https://icantech.ai/insights/graph-neural-networks-competency-dependency-modeling>
130. Design and Optimizat on of Knowledge Graph-driven Skill Mapping ... <https://dl.acm.org/doi/10.1145/3764206.3764336>
131. [PDF] TracePath: Modeling and Analyzing Competency Trajectories ... - HAL <https://hal.science/hal-05490129/document>
132. [PDF] Structuring Competency-Based Courses Through Skill Trees - arXiv <https://arxiv.org/pdf/2504.16966>
133. [PDF] Competency Based Learning Materials (CBLMs) Web Design <https://objectstorage.ap-dcc-gazipur-1.oraclecloud15.com/n/axvjbnqprylg/b/V2Ministry/o/office-nsda/2024/12/862f4eb396e546d0ba333027a201952b.pdf>
134. Competency-Based E-Learning Systems: Automated Integration of ... <https://www.mdpi.com/2071-1050/14/24/16544>

135. (PDF) Curriculum Design and Competency Mapping for ... https://www.researchgate.net/publication/395530054_Curriculum_Design_and_Competency_Mapping_for_Personalized_Learning_in_OBE
136. Developing and Designing Competency Maps Informed by ... https://edtechbooks.org/jaid_13_3/competency_maps
137. [PDF] Sequential and Repetitive Pattern Learning for Temporal ... <https://aclanthology.org/2024.lrec-main.1284.pdf>
138. Construction of a three-layer N-domain model based on knowledge ... <https://www.spiedigitallibrary.org/conference-proceedings-of-spie/14018/140184E/Construction-of-a-three-layer-N-domain-model-based-on/10.1117/12.3089415.full>
139. g3i/lxHive: A lightweight Experience API Learning Record Store (LRS) <https://github.com/g3i/lxHive>
140. THU-MAIC/OpenMAIC: Open Multi-Agent Interactive Classroom ... <https://github.com/THU-MAIC/OpenMAIC>
141. Open Source Education Platform - OpenMAIC <https://openmaic.io/openmaic-open-source-education.html>
142. Linux Foundation & CNCF Launch Free Kubernetes on Edge Training <https://training.linuxfoundation.org/blog/linux-foundation-cncf-launch-free-kubernetes-on-edge-training/>
143. 9 CNCF Training Courses That Will Help Develop Your Skills <https://vexxhost.com/blog/9-cncf-training-courses-that-will-help-develop-your-skills/>
144. Kubernetes and Cloud Native Essentials Training from CNCF and ... <https://www.youtube.com/watch?v=U05NwXkSWyU>
145. [PDF] Mathematics for Machine Learning <https://netman.aiops.org/~peidan/ANM2023/0.AIOpsBooks/mml-book.pdf>
146. Mathematics for Machine Learning [1 ed.] 110845514X ... <https://dokumen.pub/mathematics-for-machine-learning-1nbsped-110845514x-9781108455145-o-8270870.html>
147. Introduction to Linear Algebra for Applied Machine Learning with ... <https://pabloinsente.github.io/intro-linear-algebra>
148. Mathematics for Machine Learning — Theory & Implementation I <https://medium.com/@atulit23/mathematics-for-machine-learning-theory-implementation-i-d7e3b1815e4b>
149. Mathematics for Machine Learning - Google Books https://books.google.com/books/about/Mathematics_for_Machine_Learning.html?hl=id&id=pFjPDwAAQBAJ

150. A comprehensive self-study guide for the math behind machine ... https://www.reddit.com/r/learnmachinelearning/comments/ggpzk2/a_comprehensive_selfstudy_guide_for_the_math/
151. Practical Deep Learning for Coders 2022 - Fast.ai <https://www.fast.ai/posts/2022-07-21-dl-coders-22.html>
152. Practical Deep Learning for Coders 2018 - Fast.ai <https://www.fast.ai/posts/2018-01-26-v2-launch.html>
153. Practical Deep Learning for Coders (Review) <https://www.machinelearningmastery.com/practical-deep-learning-for-coders-review/>
154. Free Practical Deep Learning for Coders Course by fast.ai <https://careercoachs.com/free-courses/fastai-practical-deep-learning-for-coders>
155. The Best Machine Learning Courses to Take in 2026 | DataCamp <https://www.datacamp.com/blog/best-machine-learning-courses>
156. Efficiency and effectiveness of adaptive learning paths <https://www.emerald.com/itse/article/doi/10.1108/ITSE-12-2025-0370/1363782/Efficiency-and-effectiveness-of-adaptive-learning>
157. The Next Generation for Personalized Learning - 1EdTech <https://www.imsglobal.org/adaptive-adaptable-next-generation-personalized-learning>
158. Implementing Competency-Based Education with Learning Plans ... <https://www.mdpi.com/2076-3417/16/8/3854>
159. Intelligent tutoring in dynamic domains: a graph-based system for ... <https://link.springer.com/article/10.1007/s11423-026-10639-6>
160. Adaptive learning path generation in higher education using ... <https://journals.sagepub.com/doi/10.1177/10554181261461016>
161. [PDF] ADAPTIVE LEARNING INTERVENTIONS IN HIGHER EDUCATION <https://www.etamu.edu/wp-content/uploads/2025/09/2025-8-tagged-final-1.pdf>
162. AI-Driven Dynamic Competency Graph Model for Bridging Industry ... <https://dl.acm.org/doi/full/10.1145/3775073.3775102>
163. Fostering AI Competence as a Future Skill with AICompAss in ... <https://eurodljournal.com/en/articles/10.65043/eurodl.178>
164. [PDF] A Cross-View Hierarchical Graph Learning Hypernetwork for Skill ... <https://ojs.aaai.org/index.php/AAAI/article/view/29956/31672>
165. SkillNet: Create, Evaluate, and Connect AI Skills - arXiv <https://arxiv.org/html/2603.04448v1>
166. Designing a Competency-Focused Course on Applied AI Based on ... <https://www.mdpi.com/2076-3417/14/10/4107>

167. AI Skills Opportunity Map - Digital Education Council <https://www.digitaleducationcouncil.com/resource-library-items/ai-skills-opportunity-map>
168. (PDF) Utilizing Artificial Intelligence for Competency Mapping and ... https://www.researchgate.net/publication/389584513_Utilizing_Artificial_Intelligence_for_Competency_Mapping_and_Personalised_Skill_Development_in_IT_Organizations
169. Be AIware! An AI Competency Model for K-12 Education <https://www.sciencedirect.com/science/article/pii/S2590291125005662>
170. An intelligent curriculum alignment model for digital skills integration ... <https://link.springer.com/article/10.1007/s44217-026-01180-y>
171. Google AI Professional Certificate | Coursera <https://www.coursera.org/professional-certificates/google-ai>
172. Google Skills <https://www.skills.google/>
173. Google Skills AI Courses: What to Take in 2026 <https://spectrumailab.com/blog/google-skills-13-ai-courses-ranked-2026>
174. 7 Essential Google Courses On Artificial Intelligence in 2026 <https://mammothclub.com/blog/google-courses-on-artificial-intelligence>
175. Google AI Professional Certificate | Become Fluent in AI at Work https://grow.google/ai-professional/?srsltid=AfmBOorNxuY_lmx7VpCdEYCOPIob_5duSbm1LTnsro9XPIhAxGsG4Py6
176. Machine Learning & AI Courses | Google Cloud Training <https://cloud.google.com/learn/training/machinelearning-ai>
177. Google AI Certifications in 2026: Best AI & Cloud ... - NetCom Learning <https://www.netcomlearning.com/blog/breaking-into-ai-and-machine-learning-essential-skills-and-google-cloud-ai-certifications>
178. Learning Paths & Certifications Prep | AI Skill Pathways Library <https://learning.linkedin.com/resources/upskilling-and-reskilling/ai-skill-pathways-library>
179. Practical AI skills for everyone - Google Blog <https://blog.google/company-news/outreach-and-initiatives/grow-with-google/google-ai-professional-certificate/>
180. OpenMAIC — Open Multi-Agent Interactive Classroom by Tsinghua ... <https://openmaic.io/>
181. CBE Starter Pack 4: Progress Based on Mastery - FullScale <https://www.fullscalelearning.org/cw/cbe-starter-pack-4-progress-based-on-mastery/>
182. Releases · harvard-edge/cs249r_book - GitHub https://github.com/harvard-edge/cs249r_book/releases
183. harvard-edge/cs249r_book: Machine Learning Systems Engineering https://hellogithub.com/en/repository/harvard-edge/cs249r_book

184. CNCF Kubernetes and Cloud Native Associate Certification Course ... <https://www.youtube.com/watch?v=AplluksKvzI>
185. Khan Academy | Free Online Courses, Lessons & Practice <https://www.khanacademy.org/>
186. Introduction to AI and Machine Learning (video) | Khan Academy <https://www.khanacademy.org/partner-content/yuva-ai-for-all/x11a06d2ab0cc5322:yuva-ai/x11a06d2ab0cc5322:module-1/v/introduction-to-ai-and-machine-learning>
187. Online Course AI Notes: Video Lessons to Review - Thetawave AI <https://thetawave.ai/use-case/for-online-learners>
188. Khan Academy - OpenAI <https://openai.com/index/khan-academy/>
189. Sal and his son take the newest Open AI model for a test drive <https://www.facebook.com/khanacademy/videos/sal-and-his-son-take-the-newest-open-ai-model-for-a-test-drive/448979721016147/>
190. AI for education | Life skills - Khan Academy <https://www.khanacademy.org/college-careers-more/ai-for-education>
191. Meet Khanmigo: Khan Academy's AI-powered teaching assistant ... <https://www.khanmigo.ai/>
192. I scraped and ranked AI courses, here are the best I found - Reddit https://www.reddit.com/r/learnmachinelearning/comments/1d0bksx/i_scraped_and_ranked_ai_courses_here_are_the_best/
193. Constructing an ontology-based knowledge graph for K-12 ... <https://pure.ecnu.edu.cn/en/publications/constructing-an-ontology-based-knowledge-graph-for-k-12-computer-/>
194. Ontological approach for competency-based curriculum analysis <https://www.sciencedirect.com/science/article/pii/S2405844024050771>
195. The Competency Ontology - ResearchGate https://www.researchgate.net/figure/The-Competency-Ontology_fig1_228640284
196. [PDF] A Competency-Based Ontology for Learning Design Repositories https://thesai.org/Downloads/Volume5No1/Paper_8-A_Competency_Based_Ontology_for_Learning_Design_Repositories.pdf
197. [PDF] integration of competencies in an ontology-based framework ... <https://reference-global.com/download/article/10.2478/v10099-011-0010-4.pdf>
198. [PDF] Domain ontology for mapping competency development in higher ... <https://ceur-ws.org/Vol-4177/paper19.pdf>
199. [PDF] A new competency ontology for learning environments personalization <https://scispace.com/pdf/a-new-competency-ontology-for-learning-environments-59yov6uwcg.pdf>

200. Neural-Symbolic Knowledge Tracing: Injecting Educational ... - arXiv <https://arxiv.org/html/2604.08263v1>
201. Adaptive Curriculum Strategies: Stabilizing Reinforcement Learning... <https://openreview.net/forum?id=RGHxlzhQLN>
202. Adaptive Learning AI: Unlocking Personalized Student Paths (2026 ... <https://agixtech.com/insights/adaptive-learning-ai-personalized-paths-for-every-student/>
203. [PDF] Adaptive Curriculum Learning - CVF Open Access https://openaccess.thecvf.com/content/ICCV2021/papers/Kong_Adaptive_Curriculum_Learning_ICCV_2021_paper.pdf
204. [PDF] AdaCuRL: Adaptive Curriculum Reinforcement Learning with Invalid ... <https://ojs.aaai.org/index.php/AAAI/article/view/39479/43440>
205. Ethics-Aware AI Agents for Adaptive Education: A Multi ... - MDPI <https://www.mdpi.com/2227-7080/14/5/311>
206. What's New in Evolve: Smarter Logic, Adaptive Learning - YouTube <https://www.youtube.com/watch?v=bPosh2Bn5MM>
207. (PDF) Game Mechanics and Artificial Intelligence Personalization https://www.researchgate.net/publication/389444489_Game_Mechanics_and_Artificial_Intelligence_Personalization_A_Framework_for_Adaptive_Learning_Systems
208. Research on Algorithm Design of Intelligent Teaching System Based ... <https://dl.acm.org/doi/pdf/10.1145/3746469.3746505>
209. Adaptive learning platforms - Skillwell <https://www.skillwell.com/resources/adaptive-learning-platforms>
210. Deep Learning Foundations Signup, Open Source Scholarships ... <https://www.fast.ai/posts/2022-09-23-part2-2022-signup.html>
211. [PDF] Empowering Learners for the Age of AI | OECD https://www.oecd.org/content/dam/oecd/en/publications/reports/2026/06/empowering-learners-for-the-age-of-ai_2f8315e7/65cd27d4-en.pdf
212. Computational thinking and AI literacy integration in mathematics ... <https://link.springer.com/article/10.1007/s11858-026-01795-3>
213. Standards Alignment – Binary Logic – K-12 CS/AI Curricula <https://binarylogic.net/pedagogy/standards-alignment/>
214. Artificial Intelligence (AI) Literacy in Education - Preprints.org <https://www.preprints.org/manuscript/202508.0497>
215. Qingsong Wen's Homepage - AI for Education - Google Sites <https://sites.google.com/site/qingsongwen8/ai-for-education>

216. Artificial intelligence and education and skills | OECD <https://www.oecd.org/en/topics/artificial-intelligence-and-education-and-skills.html>
217. AI for Skills Development in Higher Education (2025–2026) - UNESCO <https://www.unesco.org/en/node/217106>
218. AI Learning Priorities for All K-12 Students <https://csteachers.org/ai-priorities/>
219. Computer Science-Integrated AI Education in K-12 <https://dl.acm.org/doi/full/10.1145/3775073.3775103>
220. Google AI Professional Certificate | Become Fluent in AI at Work <https://grow.google/ai-professional/?srsltid=AfmBOooEcTBmchaT1VzsXesNfN8innPkvQDl33sg6jdy78qRoJHBbjQO>
221. FREE Google AI Courses with Certificates in 2026! <https://www.facebook.com/groups/researchupdates.in/posts/26441422258854871/>
222. Free AI And Tech Courses With Certificates | How To Get Started ... <https://www.instagram.com/reel/DXB05FXiv7G/>
223. Demystifying Skill Ontologies: Your Roadmap to Clarity - Phenom <https://www.phenom.com/blog/demystifying-skill-ontologies>
224. Skills ontology framework: Why You need it in 2026 - Gloat <https://gloat.com/blog/skills-ontology-framework/>
225. A new competency ontology for learning environments personalization <https://link.springer.com/article/10.1186/s40561-021-00160-z>
226. Katariina Kari's Post - LinkedIn https://www.linkedin.com/posts/katsi_competency-questions-are-used-in-ontology-activity-7402953679991013376-Xc-h
227. 5 AI Layers Powering Skills Intelligence & Workforce Planning <https://www.techwolf.ai/resources/blog/inside-techwolfs-ai-engine-the-5-layers-driving-skills-intelligence>
228. Skill Data Dictionary Part 2: Organization - Degreed <https://degreed.com/experience/blog/skill-data-dictionary-part-2-organization/>
229. Skills Ontology: Definition & 6 Best Practices - 360Learning <https://360learning.com/blog/skills-ontology/>
230. Going Meta S03E07 – Agent Skills For Creating and Using Ontologies <https://www.youtube.com/watch?v=6bIUwMrKDlk>
231. So you are interested in deep learning - Fast.ai <https://www.fast.ai/posts/2016-12-19-favorite-posts.html>
232. What is better to use: MIT OCW or Khan Academy? - Quora <https://www.quora.com/What-is-better-to-use-MIT-OCW-or-Khan-Academy>
233. khan academy - Fast Company <https://www.fastcompany.com/section/khan-academy>

234. Math for ML: Your roadmap to success Machine learning ... - Instagram <https://www.instagram.com/p/CxmYpoTscjp/>
235. How Machines Learn | YUVA AI for All | MEITY | Khan Academy <https://www.youtube.com/watch?v=f-i8tJyAsns>
236. Introducing Khan Academy: Computer Science, a project led by ... https://www.reddit.com/r/programming/comments/y7ugl/introducing_khan_academy_computer_science_a/
237. ayyucedemirbas/Machine-Learning-Pathway - GitHub <https://github.com/ayyucedemirbas/Machine-Learning-Pathway>
238. Does Practice Make Perfect? Analyzing the Relationship Between ... <https://educationaldatamining.org/edm2022/proceedings/2022.EDM-short-papers.27/index.html>
239. Illustration of where feedback, support, micro-sequencing, and... https://www.researchgate.net/figure/Illustration-of-where-feedback-support-micro-sequencing-and-macro-sequencing-occur_fig2_281901125
240. Optimizing Mastery Learning by Fast-Forwarding Over-Practice Steps <https://arxiv.org/html/2506.17577v1>
241. [PDF] Adaptive AI-Driven Learning Systems for Personalized Student ... <https://egarp.lt/index.php/LUMIN/article/download/504/486>
242. [PDF] Dynamic Knowledge Tracing Models for Large-Scale Adaptive ... https://personales.upv.es/thinkmind/dl/journals/intsys/intsys_v12_n12_2019/intsys_v12_n12_2019_9.pdf
243. How mastery levels affect future learning - LinkedIn https://www.linkedin.com/posts/ryan-baker-36567482_proceedings-template-word-activity-7335703514058252290-KHPP
244. AI-Enabled Personalized Learning Path Optimization for Smart ... <https://dl.acm.org/doi/10.1145/3799457.3799519>
245. [PDF] An LLM-Powered Adaptive Practicing System - CEUR-WS.org <https://ceur-ws.org/Vol-3487/paper1.pdf>
246. Ai Certification Google Guide: Your Essential Path for 2026 <https://mammothclub.com/blog/ai-certification-google>
247. Google AI Professional Certificate <https://www.skills.google/paths/3644>
248. Google AI Professional Certificate | Become Fluent in AI at Work https://grow.google/ai-professional/?srsltid=AfmBOopxr25k-gaOlzKeEjmYEUzWms0Rd8RGlUekJxLvFPRaMFImk_ct
249. AI Professional Roadmap 2026 | AI Course For Beginners - YouTube https://www.youtube.com/watch?v=P_EBIA3Mywg

250. Machine Learning, Gamification, and Critical Thinking in Adaptive ... <https://www.mdpi.com/2078-2489/17/7/619>
251. Development and validation of an adaptive learning readiness ... https://www.researchgate.net/publication/407109821_Development_and_validation_of_an_adaptive_learning_readiness_assessment_framework_for_Moodle_courses
252. Adversarial Self-Play with Evolving Distractions for LLM Reasoning <https://arxiv.org/html/2605.11636v1>
253. [PDF] Ryan Shaun Baker - Penn Center for Learning Analytics <https://learninganalytics.upenn.edu/ryanbaker/CV.pdf>
254. [PDF] Journal of Language Teaching and Research Contents <https://www.academypublication.com/issues3/jltr/vol16/jltr1606.pdf>
255. Analysis of factors affecting the academic performance of university ... <https://www.nature.com/articles/s41598-025-28870-1>
256. Geoff Webb: Publications <https://i.giwebb.com/publications/>
257. [PDF] Studies in Educational Evaluation - selfdeterminationtheory.org https://selfdeterminationtheory.org/wp-content/uploads/2026/03/2026_GraciaCarzorlaAbosEtaL_Teaching.pdf
258. [PDF] BOOK OF ABSTRACTS - 2026 ESSE Conference https://www.esse2026.com/media/programa/ESSE%202026_Provisional%20Programme%20&%20Book%20of%20Abstracts.pdf
259. Khan Academy <https://www.khanacademy.org/computer-programming/first-program/3949243511>
260. h9-tec/AI-roadmap - GitHub <https://github.com/h9-tec/AI-roadmap>
261. Khan Academy Analytics | PPTX <https://www.slideshare.net/slideshow/khan-academy-analytics/84434366>
262. SkillNet: Create, Evaluate, and Connect AI Skills - arXiv <https://arxiv.org/html/2603.04448>
263. LLM-Powered Construction of Course Knowledge-Competency ... <https://dl.acm.org/doi/10.1145/3766557.3766569>
264. [PDF] Modeling of Skills, Abilities, Competences, and Related Dispositions ... <https://ceur-ws.org/Vol-4176/oss-4.pdf>
265. Application for Creating and Analyzing Competency-Based ... <https://ieeexplore.ieee.org/iel8/6287639/10380310/10580886.pdf>
266. [PDF] AI and the Future of Skills, Volume 1 | OECD https://www.oecd.org/content/dam/oecd/en/publications/reports/2021/11/ai-and-the-future-of-skills-volume-1_2f19d213/5ee71f34-en.pdf

267. The effects of artificial intelligence applications in educational settings <https://dx.doi.org/https://doi.org/10.1016/j.techfore.2023.123076>
268. A Survey of Agent Skills for Foundation-Model Agents - Preprints.org <https://www.preprints.org/manuscript/202604.1817>
269. Research on intelligent matching of students' learning ability and ... <https://www.frontiersin.org/journals/artificial-intelligence/articles/10.3389/frai.2025.1650095/full>
270. [PDF] AISAO: An Ontology-Hybrid AI Framework for ... - SciTePress <http://www.scitepress.org/Papers/2026/149742/149742.pdf>
271. A Multi-View Predictive Student Modeling Framework with ... <https://educationaldatamining.org/EDM2025/proceedings/2025.EDM.long-papers.135/index.html>
272. The new version of Fast.ai's Practical Deep Learning for Coders is ... https://www.reddit.com/r/learnmachinelearning/comments/w4x626/the_new_version_of_fastais_practical_deep/
273. 从深度学习到扩散模型的手把手入门实战中文教程Practical Deep ... <https://event.baai.ac.cn/activities/759>
274. [PDF] Empowering Learners for the Age of AI - AILit Framework https://ailiteracyframework.org/wp-content/uploads/2025/05/AILitFramework_ReviewDraft.pdf
275. What should teachers teach and students learn in a future of ... - OECD https://www.oecd.org/en/publications/what-should-teachers-teach-and-students-learn-in-a-future-of-powerful-ai_ca56c7d6-en.html
276. CNCF, Your Certification Exams Are a Privileged, Ableist Joke - Reddit https://www.reddit.com/r/devops/comments/1l88uej/cncf_your_certification_exams_are_a_privileged/
277. Cloud Native Certifications | CNCF <https://www.cncf.io/training/certification/>
278. Computer Science Education - SIGCSE 2024 - ACM Digital Library <https://dl.acm.org/doi/proceedings/10.1145/3626252?tocHeading=heading2>
279. Machine Learning, Gamification, and Critical Thinking in Adaptive ... <https://doi.org/10.3390/info17070619>
280. [PDF] Evaluating and Improving Teacher Preparation Programs https://naeducation.org/wp-content/uploads/2025/01/Evaluating_and_Improving_Teacher_Preparation_Programs.pdf
281. [PDF] Table of content - Diva-Portal.org <https://www.diva-portal.org/smash/get/diva2:1920905/FULLTEXT01.pdf>
282. [PDF] TUW-Catalog-2426.pdf <https://www.tuw.edu/wp-content/uploads/TUW-Catalog-2426.pdf>

283. [PDF] 2024-25 Bulletin - Grove City College <https://www.gcc.edu/Portals/0/2024-25-Catalog.pdf>
284. [PDF] Education and New Developments 2025 - Volume 1 https://end-educationconference.org/wp-content/uploads/2025/09/Education-and-New-Developments_2025-_Vol_I.pdf
285. [PDF] Untitled - PhilPapers <https://philpapers.org/archive/LEYTHC.pdf>
286. [PDF] Poverty in Children and Youth-Interventions and Context - ISSBD https://issbd.org/wp-content/uploads/2025/12/JBD_49_6S_Combined-PDF.pdf
287. [PDF] IU Northwest Bulletin 2024-26 <https://bulletins.iu.edu/iun/2024-2026/iun-pdf.pdf>
288. AI-Powered Curriculum Mapping: Revolutionizing Academic ... <https://www.mapademics.com/resources/whitepapers/ai-curriculum-mapping-program-design>
289. Integrating AI and Learning Analytics for Data-Driven Pedagogical ... <https://link.springer.com/article/10.1007/s10758-025-09897-9>
290. [PDF] mapping insights <https://ai-teach.d-teachtraining.com/wp-content/uploads/2025/12/Mapping-Insights-Final-Report.pdf>
291. A Framework for Generative AI-Driven Assessment in Higher ... - MDPI <https://www.mdpi.com/2078-2489/16/6/472>
292. Automating the Mapping of Course Learning Outcomes to Program ... https://www.researchgate.net/publication/364707363_Automating_the_Mapping_of_Course_Learning_Outcomes_to_Program_Learning_Outcomes_using_Natural_Language_Processing_for_Accurate_Educational_Program_Evaluation
293. A systematic review on the future of educational assessment: AI ... <https://www.emerald.com/aiie/article/2/2/75/1317167/A-systematic-review-on-the-future-of-educational>
294. Artificial intelligence-enabled adaptive learning platforms: A review <https://www.sciencedirect.com/science/article/pii/S2666920X25000694>
295. Preparing Students for AI-Powered Materials Discovery - arXiv <https://arxiv.org/html/2605.09624v1>
296. Artificial Intelligence in Medical Education: Transforming Learning ... <https://pmc.ncbi.nlm.nih.gov/articles/PMC12007958/>
297. AI Curriculum Mapping Generator | Taskade AI <https://www.taskade.com/generate/content/curriculum-mapping>
298. Free MIT, Stanford, and Carnegie Mellon CS Curriculum on GitHub ... <https://www.instagram.com/reel/DXpGxTSCIUN/>

299. sannykim/deeplearning-guide: An evolving guide to learning Deep ... <https://github.com/sannykim/deeplearning-guide>
300. GitHub - WA9ONY/AI: Artificial intelligence notes <https://github.com/WA9ONY/AI>
301. Khan/khan-exercises: A (deprecated) framework for ... - GitHub <https://github.com/Khan/khan-exercises>
302. From Zero to AI Research Scientist Full Resources Guide - Facebook <https://www.facebook.com/groups/MontrealAI/posts/1633777833750709/>
303. New program - Khan Academy <https://www.khanacademy.org/computer-programming/new-program/4610389275164672>
304. ashleymcnamara/learn_to_code: Resources for budding developers https://github.com/ashleymcnamara/learn_to_code
305. Github Repositories To Learn Data Science And Generative AI With ... <https://www.youtube.com/watch?v=CXJo-XWBRPs>
306. Practical Deep Learning - Lesson 1 - Exploring ML <https://exploringml.com/posts/practical-deep-learning-for-coders-lesson-1/>
307. The Ultimate Self-Learner's Roadmap to Mastering AI in 2026 (For ... <https://medium.com/@rickblosser/the-ultimate-self-learners-roadmap-to-mastering-ai-in-2026-for-free-5fc311bc9a95>
308. Google AI Professional Certificate | Become Fluent in AI at Work https://grow.google/ai-professional/?srsltid=AfmBOorn10cKULRjB9amxnH4HxPFOFxU-XZuG6cOwW2bYCDUCV_n1Keh
309. 7 Google AI Certifications to Boost Your Salary (Beginner ... - YouTube <https://www.youtube.com/watch?v=wH-09620ntU>
310. Best AI Certifications for 2026 (Ranked & Compared) - Dataquest <https://www.dataquest.io/blog/best-ai-certifications/>
311. Mastery levels and mastery goals on Khan Academy (video) <https://www.khanacademy.org/khan-for-educators/khan-for-educators-advanced-course/x2e5750eab575b791:khan-for-educators-advanced/x2e5750eab575b791:mastery-learning-on-khan-academy/v/mastery-levels-and-mastery-goals-on-khan-academy>
312. How Khan Academy Became a Leader in A/B Testing AI ... - YouTube <https://www.youtube.com/watch?v=AzAeirxwPJk>
313. Mastery Learning on Khan Academy (article) <https://en.khanacademy.org/khan-for-educators/khan-for-educators-advanced-course/x2e5750eab575b791:khan-for-educators-advanced/x2e5750eab575b791:mastery-learning-on-khan-academy/a/mastery-learning-on-khan-academy>
314. Part 5: Bringing AI to the classroom (video) - Khan Academy <https://www.khanacademy.org/college-careers-more/ai-for-education/>

x68ea37461197a514:unit-teaching-with-ai/x68ea37461197a514:ai-101-for-teachers-lesson/v/ai-101-for-teachers-bringing-ai-to-the-classroom

315. How Khan Academy is Bringing Mastery Learning to the Masses <https://www.cultofpedagogy.com/khan-mastery-learning/>
316. How Khan Academy optimizes AI tutoring with experimentation <https://www.growthbook.io/blog/how-khan-academy-optimizes-ai-tutoring-with-experimentation>
317. Khan Academy view of mastery learning - YouTube <https://www.youtube.com/watch?v=GWa48XRnLh0>
318. How to assign course mastery goals? (article) - Khan Academy <https://en.khanacademy.org/khan-for-educators/khan-for-educators-advanced-course/x2e5750eab575b791:khan-for-educators-advanced/x2e5750eab575b791:mastery-learning-on-khan-academy/a/how-to-assign-mastery-goals-a9>
319. dmccreary/claude-skills - GitHub <https://github.com/dmccreary/claude-skills>
320. AI Research Skills Library - GitHub <https://github.com/orchestra-research/AI-research-SKILLs>
321. VoltAgent/awesome-agent-skills: A curated collection of ... - GitHub <https://github.com/VoltAgent/awesome-agent-skills>
322. Building Intelligent Textbooks With AI - GitHub <https://github.com/dmccreary/intelligent-textbooks>
323. spcl/graph-of-thoughts: Official Implementation of "Graph of ... - GitHub <https://github.com/spcl/graph-of-thoughts>
324. kodustech/awesome-agent-skills - GitHub <https://github.com/kodustech/awesome-agent-skills>
325. Egonex-AI/Understand-Anything - GitHub <https://github.com/Egonex-AI/Understand-Anything>
326. managedcode/dotnet-skills: Installable .NET skill catalog ... - GitHub <https://github.com/managedcode/dotnet-skills>
327. AbhishekSharma-17/skills-graph: A collection of agents ... - GitHub <https://github.com/abhisheksharma-17/skills-graph>
328. K-Dense-AI/scientific-agent-skills - GitHub <https://github.com/k-dense-ai/scientific-agent-skills>
329. [PDF] Digital Intelligence Education - DI-IDEA - 北京大学 <https://diidea.pku.edu.cn/userfiles/resources/202601121432501036.pdf>
330. AI-Powered Curriculum Mapping for Academic Excellence <https://www.mapademics.com/resources/whitepapers/ai-powered-curriculum-mapping-for-academic-excellence>

331. A Comparative Analysis of On-Device AI-Driven, Self-Regulated ... https://www.researchgate.net/publication/388889101_A_Comparative_Analysis_of_On-Device_AI-Driven_Self-Regulated_Learning_and_Traditional_Pedagogy_in_University_Health_Sciences_Education
332. From principles to practice: a novel matrix for evaluating AI-powered ... <https://www.frontiersin.org/journals/education/articles/10.3389/feduc.2025.1640780/full>
333. Research on Learning Analytics–Driven AI-Supported Blended ... <https://www.mdpi.com/2227-7102/16/6/876>
334. A research roadmap for AI opportunities in student assessment for ... <https://pmc.ncbi.nlm.nih.gov/articles/PMC12659470/>
335. Best AI tool for online course mapping and alignment? - Facebook <https://www.facebook.com/groups/703007927897194/posts/832885934909392/>
336. Google AI Professional Certificate | Become Fluent in AI at Work https://grow.google/ai-professional/?srsltid=AfmBOoqDBHRTCKDVZiXreo3H3E2eRASz3t3_BwHA2gju7pesUoHsUhQL
337. HazyResearch/skill-it: Skill-It! A Data-Driven Skills ... - GitHub <https://github.com/HazyResearch/skill-it>
338. K-Dense-AI/scientific-agent-skills - GitHub <https://github.com/K-Dense-AI/scientific-agent-skills>
339. agents/docs/agent-skills.md at main · wshobson/agents - GitHub <https://github.com/wshobson/agents/blob/main/docs/agent-skills.md>
340. An Open Source Machine Learning Framework for Everyone · GitHub <https://github.com/tensorflow/tensorflow>
341. Meet GitNexus: An Open-Source MCP-Native Knowledge Graph ... <https://www.marktechpost.com/2026/04/24/meet-gitnexus-an-open-source-mcp-native-knowledge-graph-engine-that-gives-claude-code-and-cursor-full-codebase-structural-awareness/>
342. product-on-purpose/pm-skills: 67 plug-and-play, best ... - GitHub <https://github.com/product-on-purpose/pm-skills>
343. I Found the Best Claude Skills GitHub Repos — Here's What's ... <https://medium.com/all-about-claude/i-found-the-best-claude-skills-github-repos-heres-what-s-actually-worth-installing-in-2026-506aacd22ee5>
344. 5 Claude Code skills I use every single day - YouTube <https://www.youtube.com/watch?v=EJyuu6zlQCg>
345. Building AI Companions that Prioritise Learning over Performance https://www.researchgate.net/publication/404542251_Building_AI_Companions_that_Prioritise_Learning_over_Performance

346. The Quiet Math of EdTech — Can AI Tutors Really Teach? - Medium <https://medium.com/@adnanmasood/the-quiet-math-of-edtech-can-ai-tutors-really-teach-737195005abf>
347. [PDF] Student-Centered Learning: https://aurora-institute.org/wp-content/uploads/iNACOL_FunctionalRequirementsForIntegratedSystems.pdf
348. [PDF] Building AI Companions that Prioritise Learning over Performance <https://arxiv.org/pdf/2605.04816>
349. Challenging Cognitive Load Theory: The Role of Educational ... - PMC <https://pmc.ncbi.nlm.nih.gov/articles/PMC11852728/>
350. THE BROKEN PROMISES OF ED TECH - Facebook <https://www.facebook.com/groups/279950626943/posts/10164865737531944/>
351. [PDF] Game Architecture in Transmedia Education (GATE) <https://journal.seriousgamessociety.org/index.php/IJSG/article/download/1115/669/7656>
352. [PDF] Handbook of Learning Analytics <https://solaresearch.org/wp-content/uploads/2017/05/hla17.pdf>
353. [PDF] AN INTERNATIONAL PERSPECTIVE ON DIGITAL LITERACY - IEA.nl https://www.iea.nl/sites/default/files/2024-11/ICILS_2023_International_Report_0.pdf
354. [PDF] Assessment Framework for the Digital SAT Suite <https://web.ped.nm.gov/wp-content/uploads/2025/01/assessment-framework-for-digital-sat-suite.pdf>
355. Leveraging AI in E-Learning: Personalized Learning and Adaptive ... <https://www.mdpi.com/2079-9292/13/18/3762>
356. AI Video Analytics for Online Learning in 2026 - Fora Soft <https://www.forasoft.com/blog/article/ai-video-analytics-online-learning>
357. Transforming EFL Teaching with AI: A Systematic Review of ... <https://www.sciencedirect.com/science/article/pii/S1560429226000375>
358. The Top 11 AI-Powered Learning Platforms in 2026 - 360Learning <https://360learning.com/blog/ai-learning-platforms/>
359. Powerful Tools and Techniques for AI in Formative Assessment <https://thecasehq.com/powerful-tools-and-techniques-for-ai-in-formative-assessment/?srsltid=AfmBOoqlsh83VRO53Wp2L2fGOsIvWEA1KtwHB6kjFLBcBq6L-HKdfAny>
360. New AI Literacy Framework to Equip Youth in an Age of AI <https://oecdeditoday.com/new-ai-literacy-framework-to-equip-youth-in-an-age-of-ai/>
361. AI Literacy Framework for Primary & Secondary Education— OECD ... <https://learnworkecosystemlibrary.com/initiatives/ai-literacy-framework-for-primary-secondary-education-oecd-ec/>
362. OECD Digital Education Outlook 2026 https://www.oecd.org/en/publications/oecd-digital-education-outlook-2026_062a7394-en.html

363. OECD Digital Education Outlook 2026: how generative AI can ... <https://digital-skills-jobs.europa.eu/en/latest/news/oecd-digital-education-outlook-2026-how-generative-ai-can-support-learning-when-used>
364. OECD, UNESCO, and EU Governance Frameworks Training Course <https://upskilldevelopment.com/global-standards-for-ai-and-ict-oecd-unesco-and-eu-governance-frameworks-training-course>
365. The Future of AI Literacy Education: Integrating OECD Education ... https://www.researchgate.net/publication/392245768_The_Future_of_AI_Literacy_Education_Integrating_OECD_Education_2030_with_Technical_and_Ethical_Perspectives
366. Google AI Professional Certificate | Become Fluent in AI at Work https://grow.google/ai-professional/?srsltid=AfmBOor_ePyQ733SiGJwPralEjEUC_gColSR-Pp4I2LAdDp94Sjv1w
367. Raising kids who can't be automated in K-12 education - Facebook <https://www.facebook.com/groups/181538007171095/posts/raising-kids-who-can-t-automated-k-12-education/1333951515263066/>
368. [PDF] Moving Measurement Forward - NCME https://ncme.org/wp-content/uploads/2026/04/2026_NCMEprogram_04072026v2.pdf
369. [PDF] Design Recommendations for Intelligent Tutoring Systems https://gifttutoring.org/attachments/download/3410/DesignRecommendationsforITS_Volume7_SelfImprovingSystemsBook.pdf
370. josephmisiti/awesome-machine-learning: A curated list of ... - GitHub <https://github.com/josephmisiti/awesome-machine-learning>
371. mikeroyal/Machine-Learning-Guide - GitHub <https://github.com/mikeroyal/Machine-Learning-Guide>
372. dair-ai/ML-YouTube-Courses: Discover the latest machine learning ... <https://github.com/dair-ai/ML-YouTube-Courses>
373. Category Theory $\cap$ Machine Learning - GitHub https://github.com/bgavran/Category_Theory_Machine_Learning
374. GitHub - rahulnyk/knowledge_graph: Convert any text to a graph of ... https://github.com/rahulnyk/knowledge_graph
375. dair-ai/Prompt-Engineering-Guide - GitHub <https://github.com/dair-ai/prompt-engineering-guide>
376. Adaptive learning featured in HarvardX course - Harvard Gazette <https://news.harvard.edu/gazette/story/2017/02/adaptive-learning-featured-in-harvardx-course/>
377. Add adaptive learning capabilities to open edX - ed | Xchange <https://edxchange.opencraft.com/t/add-adaptive-learning-capabilities-to-open-edx/80/>

378. An Evaluation of Open Source Adaptive Learning Solutions - MDPI [https://
www.mdpi.com/2078-2489/14/2/57](https://www.mdpi.com/2078-2489/14/2/57)
379. Open edX | Schema Education <https://schema.education/open-edx/>